

Research Notes on Graph neural network

Graph neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

>> Section 1

$$h_u = \sum_{k=1}^K \sigma \left(\sum_{v \in N_u} \alpha_{uv} W_k x_v \right)$$

>> Section 2

In the more general subject of "geometric deep learning", certain existing neural network architectures can be interpreted as GNNs operating on suitably defined graphs. A convolutional neural network layer, in the context of computer vision, can be considered a GNN applied to graphs whose nodes are pixels, and only adjacent pixels are connected by edges in the graph. A transformer layer, in natural language processing, can be considered a GNN applied to complete graphs whose nodes are words or tokens in a passage of natural language text. Relevant application domains for GNNs include natural language processing, social networks, citation networks, molecular biology, chemistry, physics and NP-hard combinatorial optimization problems. Open source libraries implementing GNNs include PyTorch Geometric (PyTorch), TensorFlow GNN (TensorFlow), Deep Graph Library (framework agnostic), jraph (Google JAX), and GraphNeuralNetworks.jl/GeometricFlux.jl (Julia, Flux).

>> Section 3

Gated graph sequence neural network The gated graph sequence neural network (GGS-NN) was introduced by Yujia Li et al. in 2015. The GGS-NN extends the GNN formulation by Scarselli et al. to output sequences. The message passing framework is implemented as an update rule to a gated recurrent unit (GRU) cell. A GGS-NN can be expressed as follows:

>> Section 4

where $\mathbf{||}$ denotes vector concatenation, $\mathbf{0}$ is a vector of zeros, Θ is a matrix of learnable parameters, GRU is a GRU cell, and l denotes the sequence index. In a GGS-NN, the node representations are regarded as the hidden states of a GRU cell. The initial node features $x_u^{(0)}$ are zero-padded up to the hidden state dimension of the GRU cell. The same GRU cell is used for updating representations for each node.

>> Section 5

$$h_u = \phi(x_u, \sum_{v \in N_u} \psi(x_u, x_v, e_{uv}))$$

>> Section 6

Graph attention network The graph attention network (GAT) was introduced by Petar Veličković et al. in 2018. A graph attention network is a combination of a GNN and an attention layer. The implementation of attention layer in graphical neural networks helps provide attention or focus to the important information from the data instead of focusing on the whole data. A multi-head GAT layer can be expressed as follows:

>> Section 7

Message passing layers are permutation-equivariant layers mapping a graph into an updated representation of the same graph. Formally, they can be expressed as message passing neural networks (MPNNs). Let $G = (V, E)$ be a graph, where V is the node set and E is the edge set. Let N_u be the neighbourhood of some node $u \in V$. Additionally, let x_u be the features of node $u \in V$, and e_{uv} be the features of edge $(u, v) \in E$. An MPNN layer can be expressed as follows:

>> Section 8

Attention in Machine Learning is a technique that mimics cognitive attention. In the context of learning on graphs, the attention coefficient α_{uv} measures how important node v is to node u . Normalized attention coefficients are computed as follows:

>> Section 9

GNNs are used as fundamental building blocks for several combinatorial optimization algorithms. Examples include computing shortest paths or Eulerian circuits for a given graph, deriving chip placements superior or competitive to handcrafted human solutions, and improving expert-designed branching rules in branch and bound.

>> Section 10

Permutation equivariant: a permutation equivariant layer maps a representation of a graph into an updated representation of the same graph. In the literature, permutation equivariant layers are implemented via pairwise message passing between graph nodes.

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Intuitively, in a message passing layer, nodes update their representations by aggregating the messages received from their immediate neighbours. As such, each message passing layer increases the receptive field of the GNN by one hop. Local pooling: a local pooling layer coarsens the graph via downsampling. Local pooling is used to increase the receptive field of a GNN, in a similar fashion to pooling layers in convolutional neural networks. Examples include k-nearest neighbours pooling, top-k pooling, and self-attention pooling. Global pooling: a global pooling layer, also known as readout layer, provides fixed-size representation of the whole graph. The global pooling layer must be permutation invariant, such that permutations in the ordering of graph nodes and edges do not alter the final output. Examples include element-wise sum, mean or maximum. It has been demonstrated that GNNs cannot be more expressive than the Weisfeiler–Leman Graph Isomorphism Test. In practice, this means that there exist different graph structures (e.g., molecules with the same atoms but different bonds) that cannot be distinguished by GNNs. More powerful GNNs operating on higher-dimension geometries such as simplicial complexes can be designed. As of 2022, whether or not future architectures will overcome the message passing primitive is an open research question.

>> Section 11

$$\alpha_{uv} = \exp \left(\text{LeakyReLU} \left(\mathbf{a}^T [\mathbf{W} \mathbf{x}_u \parallel \mathbf{W} \mathbf{x}_v] \right) \right) \sum_{z \in N_u} \exp \left(\text{LeakyReLU} \left(\mathbf{a}^T [\mathbf{W} \mathbf{x}_u \parallel \mathbf{W} \mathbf{x}_z] \right) \right) \left\{ \exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W} \mathbf{x}_u \parallel \mathbf{W} \mathbf{x}_v])) \sum_{z \in N_u} \exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W} \mathbf{x}_u \parallel \mathbf{W} \mathbf{x}_z])) \right\}$$

>> Section 12

Water distribution systems can be modelled as graphs, being then a straightforward application of GNN. This kind of algorithm has been applied to water demand forecasting, interconnecting District Metered Areas(DMAs) to improve the forecasting capacity. Other application of this algorithm on water distribution modelling is the development of metamodels.